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Assessing the anaerobic biodegradability of PHB filament and its 3D printed sheets through biochemical methane potential (BMP) test

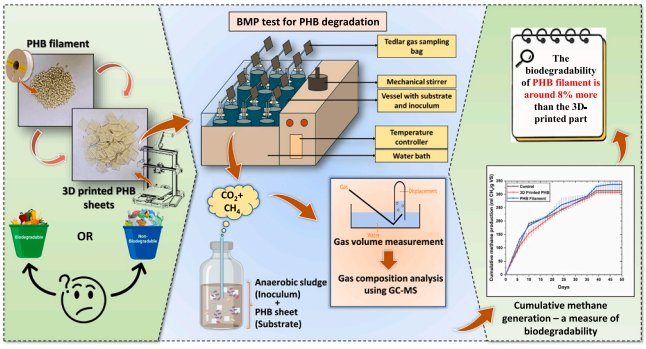
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HIGHLIGHTS

- First study to apply BMP assay for comparing biodegradation of PHB filament and 3D print.
- Processing of PHB using FDM 3D printing increases its crystallinity from 40.9 % to 61.8 %.
- BMP tests show 8 % lower anaerobic biodegradation in 3D-printed PHB (70.4 %) versus filament (78.2 %) over 50 days.
- 3D printing-induced recrystallization lowers PHB biodegradability.

GRAPHICAL ABSTRACT



ARTICLE INFO

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ABSTRACT

The incorporation of biodegradable polymers into additive manufacturing offers a promising pathway to reduce reliance on petroleum-derived materials and close the loop in circular manufacturing systems. This study aimed to evaluate how fused deposition modelling (FDM) processing affects the anaerobic biodegradability of poly(3-hydroxybutyrate) (PHB) by comparing filament and 3D-printed sheet samples. PHB samples were characterized by Fourier Transform Infrared Spectroscopy (FTIR) and X-ray Diffraction (XRD) to quantify changes in chemical bonding and crystallinity, revealing an increase from 40.9 % in the filament to 61.8 % in the 3D-printed sheet. Biochemical Methane Potential (BMP) tests, conducted at 38 °C with acclimatized anaerobic sludge (inoculum-to-substrate ratio 2.85 g VS/g VS), measured cumulative methane production over 50 days according to ASTM D5511–18 protocol. Theoretical methane yields, calculated from elemental analyses, served as benchmarks. Results showed that the filament attained 78.24 % biodegradation in 42 days (corresponding to 389 mL CH_4/g VS), whereas the 3D-printed sheet reached only 70.35 % in 49 days (equivalent to 350 mL CH_4/g VS), indicating an 8 % reduction in conversion and a 10 % lower methane yield for the printed form. This variability is attributed to processing-induced recrystallization, which limits water penetration and enzymatic access in more crystalline

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regions. To the best of our knowledge, this is the first study to determine the biodegradability of 3D-printable PHB filament and its 3D-printed counterpart using the Biochemical Methane Potential (BMP) assay. As we demonstrated, the decreased biodegradability of the 3D-printed part compared to the filament is likely due to increased crystallinity induced during processing. This insight will aid in designing material formulations and processing strategies that balance printability and end of life biodegradability, advancing the development of truly sustainable additive manufacturing solutions.

1. Introduction

In recent years, three-dimensional (3D) printing technology has emerged as an important advancement in advanced manufacturing, transforming conventional production methods through its ability to create 3D structures via layer-by-layer material deposition according to the digital designs (Gibson et al., 2010). This process, known as additive manufacturing (AM), has surpassed traditional limitations, allowing for the creation of intricate geometries that are frequently unobtainable with conventional subtractive manufacturing methods. 3D printing has applications across various industries, including aerospace, healthcare, automotive, and consumer goods, marking a significant change in the conception, design, and production of products (Abdulhameed et al., 2019).

According to ASTM 52900–21, the classification of additive manufacturing processes has defined seven distinct categories based on the mechanism and materials involved (de Pastre et al., 2022): Binder Jetting (BJT), Directed Energy Deposition (DED), Material Extrusion (MEX), Material Jetting (MJT), Powder Bed Fusion (PBF), Sheet Lamination (SHL), and Vat Photopolymerization (VPP). Each category utilizes unique principles to achieve material fusion and part formation. For example, in Binder Jetting (BJT), a liquid bonding agent is precisely deposited to join powder materials (Ziaee and Crane, 2019). Whereas Directed Energy Deposition (DED) involves the application of focused thermal energy, such as a laser or electron beam, to fuse materials by melting as they are deposited (Svetlizky et al., 2021). Material Extrusion (MEX), a category that includes fused deposition modelling (FDM), selectively dispenses material through a nozzle to construct parts layer by layer (Altıparmak et al., 2022). Powder Bed Fusion (PBF), meanwhile, applies thermal energy to selectively fuse areas of a powder bed, supporting high-resolution structures and complex geometries (Ladani and Sadeghilaridjani, 2021). Material Jetting (MJT) deposits droplets of photopolymer resin or wax, building components with fine detail (Gülcan et al., 2021), while Vat Photopolymerization (VPP) cures liquid photopolymers within a vat through light-activated polymerization (Zhang et al., 2024a). Lastly, Sheet Lamination (SHL) achieves part formation by bonding sheets of material layer by layer (Friel, 2021).

Fused Deposition Modelling (FDM) is a widely utilized and accessible 3D printing technology recognized for its ease and adaptability. The FDM technique entails the extrusion of thermoplastic filaments through a heated nozzle, whereby the material is liquefied and deposited in layers to construct the intended item (Cano-Vicent et al., 2021). This technology has numerous advantages, such as cost efficiency, user-friendliness, and the capacity to generate robust prototypes with mechanical qualities appropriate for functional testing. A key factor in the popularity of FDM is its compatibility with various thermoplastic materials, such as polylactic acid (PLA), acrylonitrile butadiene styrene (ABS), and polycarbonate (PC) (Sathies et al., 2020). There has been an increasing interest in integrating biodegradable polymers, such as polyhydroxyalkanoates (PHA), into Fused Deposition Modelling (FDM) processes. PHA, a category of biopolymer generated through microbial fermentation, offers a distinctive opportunity to tackle urgent issues of sustainability and environmental impact in production (Giubilini et al., 2021). The incorporation of PHA with FDM not only improves the environmental sustainability of printed products but also broadens the usefulness and applicability of 3D-printed objects. The integration of PHA with FDM opens new avenues for producing sustainable and functional materials, catering to the increasing demand for environmentally friendly solutions in various sectors. PHA exhibits favorable properties such as biodegradability, biocompatibility, and excellent mechanical performance, making it suitable for applications in the medical field, including tissue engineering and drug delivery systems (Gregory et al., 2023). Additionally, the versatility of PHA allows for the development of tailored blends with other polymers, thereby enhancing material properties while maintaining sustainable practices (Żur-Pińska et al., 2023).

3D printing, particularly through FDM, significantly impacts the structural and mechanical properties of materials by introducing factors like anisotropy, thermal stress, and changes in microstructure (Zohdi and Yang, 2021; Prajapati et al., 2018). The layer-by-layer construction of FDM creates directional weaknesses, often resulting in reduced strength along the vertical axis (*Z*-axis) compared to horizontal orientations. High printing temperatures can lead to thermal degradation of materials like PHA, reducing molecular weight and thus tensile strength and flexibility (Kanabenja et al., 2024). Additionally, the FDM process can produce microscopic voids, or porosity, within printed parts, compromising compressive and tensile strength. Rapid cooling can also alter the crystallinity of the polymer, impacting flexibility and stiffness (Wickramasinghe et al., 2020). These combined effects of altered microstructure, porosity, and thermal stress influence the material's mechanical performance and, for biodegradable materials like PHA, may affect degradation rates, potentially altering environmental sustainability outcomes. Therefore, understanding the biodegradability of 3D-printed specimens is a crucial area of study that can aid in identifying sustainable resources for 3D printing. For a broader context, studies on other biodegradable polymers, such as PLA and PBS, as well as PHA-based copolymers including PHBV, have shown that processing and crystallinity can strongly influence degradation rates. For example, researchers developed single-cell protein (SCP) bioplastic films from waste-derived nitrogen and carbon, producing edible-film-like materials with moderate translucency, high porosity, and low mechanical strength due to intact microbial cells (Bjurström et al., 2024). Despite their lower strength, these films exhibited excellent mesophilic anaerobic biodegradability, achieving 100 % degradation and high biomethane yields.

In this study, we aim to investigate the anaerobic biodegradability of PHB before and after processing via FDM 3D printing. Although PHAs, including PHB, are recognized for their biodegradability and environmental benefits, the impact of processing through 3D printing on their biodegradation properties remains largely unexplored. The additive manufacturing process, particularly FDM, involves thermal and mechanical stresses that could alter the material's molecular structure and, consequently, its degradation behaviour in anaerobic conditions. The Biochemical Methane Potential (BMP) test is a well-established method for evaluating the biodegradability of organic materials under anaerobic conditions by measuring their potential to generate methane during degradation by anaerobic microorganisms. By using the BMP test based on American Society for Testing and Materials (ASTM) D5511–11, this study evaluates the extent to which FDM processing affects the anaerobic biodegradability of PHB, thus providing insights into the sustainability of 3D-printed bioplastic products. Through this approach, we aim to deepen our understanding of the environmental impact and sustainability of new bioplastic formulations and how 3D printing impacts the life cycle of biodegradable polymers, informing future design and material selection for sustainable manufacturing.

2. Materials and methods

2.1. PHB procurement

The commercial PHB filament has been obtained from ColorFabb, Netherlands. The material was stored in a desiccator once it was opened so as to keep it away from moisture and other contaminants. The material properties after 3D printing, as provided by the manufacturer, are provided in Table 1.

2.2. 3D printing of PHB

Various geometries were designed using the commercial CAD software (SolidWorks) and then sliced using Ultimaker Cura and finally 3D printed using a direct drive FDM setup as shown in Fig. 1 (Creality Ender 3 S1 Pro). The direct-drive configuration enhances material control by positioning the extruder drive gears close to the hotend, thereby minimizing the filament travel distance. This is particularly important for PHB, which is a relatively brittle and highly crystalline polymer. Its stiffness and low ductility make it prone to fracture under the bending and compression stresses encountered in longer feeding paths, such as those in bowden-tube systems. Moreover, PHB has a narrow processing window between its melting point and degradation temperature, along with melt-flow properties that are sensitive to temperature fluctuations. A shorter, more direct filament path ensures smoother feeding, more consistent extrusion pressure, and reduced risk of filament breakage, ultimately improving printability and dimensional accuracy for PHB-based materials. Key printing process parameters, such as layer height, infill density, print speed, and nozzle temperature, which are crucial in achieving optimal print quality and structural integrity, are detailed in Table 2. These parameters were carefully selected to ensure high dimensional accuracy of the printed objects.

2.3. Characterization of 3D-printed PHB sheets

2.3.1. Fourier Transform Infrared Spectroscopy (FTIR)

Infrared spectra of the sample were collected using a Bruker Lumos II FTIR spectrometer. Spectra were recorded over the range of 4000–550 cm^{-1} with a resolution of 4 cm^{-1}, using the attenuated total reflectance (ATR) mode with ten scans per spectrum. During the measurement process, a Germanium (Ge) ATR crystal tip, approximately 100 μm in diameter, made contact with pre-selected focused areas on the sample. The OPUS software facilitated data analysis and visualization. Baseline correction was performed using the concave rubber band method set to 64 points, while smoothing was applied using the Savitzky–Golay filter.

2.3.2. X-ray diffraction (XRD)

XRD analysis was conducted using a PANalytical EMPYREAN system with Cu-Kα radiation, scanning at a speed of 2° per minute, to examine the crystallinity peaks in the samples. The crystallinity percentage (C (%)) was calculated by applying the peak integration function available in the Origin Pro 2022b graphing and analysis software, with the integrated values then used in the Eq. (1) for determining crystallinity.

Table 1
Material properties of 3D printed PHB filament.

Properties	Values	Testing method
Youngs Modulus (MPa)	2500	ISO 527-1 A
Tensile Strength (MPa)	26	ISO 527-1 A
Elongation at break (%)	4.5	ISO 527-1 A
Flexural Modulus (MPa)	1820	ISO 178
Flexural Strength (MPa)	41	ISO 178
Impact Strength (KJ/m^2)	3.4	ISO 179
Hardness (Shore D)	62	ISO 7619
Heat Deflection Temperature (°C)	130	ISO 75

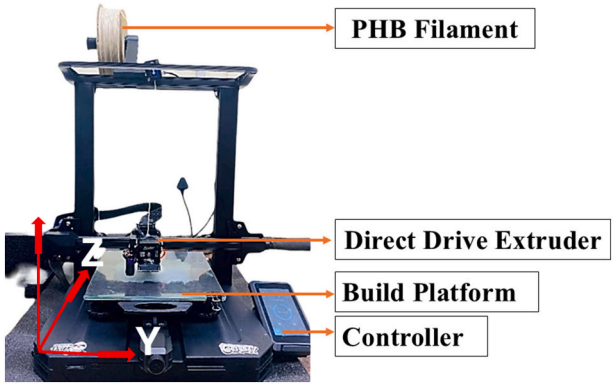


Fig. 1. FDM 3D Printing setup.

Table 2
Key Printing parameters considered for FDM 3D printing of PHB.

Factors	Values
Extruder movement speed (mm/s)	30
Layer thickness (mm)	0.12
Liquefier temperature (°C)	200
Raster angle (degree)	+45, −45
Nozzle diameter (mm)	0.4
Infill percentage (%)	20
Built platform temperature (°C)	0
Speed of retraction (mm/s)	25
First layer thickness (mm)	0.3

$$C\,(\%) = \frac{Area\ of\ crystalline\ peaks}{Area\ of\ all\ peaks\ (Crystalline + Amorphous)} \times 100 \quad (1)$$

2.4. BMP test for assessing the biodegradability

The methodology adopted for this study was in accordance with the guidelines of ASTM D5511–18, which describes the standard test method for determining anaerobic biodegradation of plastic material under high-solids anaerobic digestion conditions. (ASTM, 2012).

2.4.1. Substrates and inoculum

A commercially available PHB filament is used, and it is 3D-printed to produce sheets for subsequent testing. The filaments were cut into pieces of mean particle size 1 mm, and the sheets were cut into square pieces of size 1 cm × 1 cm. Microcrystalline cellulose powder (purchased from Sigma Aldrich) was employed as the positive control. The capacity of the inoculum used in this study to convert organic compounds into methane was consistently monitored by conducting the BMP test on this control. The anoxic sludge collected from the wastewater treatment plant of the Indian Institute of Technology, Kharagpur, was used as the inoculum for the test. The sludge was maintained under mesophilic temperature (37 °C) and anaerobic conditions for 30 days for acclimatization before its use in the BMP test. The inoculum was fed with glucose solution twice a week to ensure bacterial growth during the acclimatization period. The inoculum was sieved through 2 mm to remove the non-degraded organic fraction and thus reduce the endogenous methane production. The initial characteristics of the inoculum were analyzed to ensure its quality.

2.4.2. Analytical methods

The characterization of the PHB samples, cellulose, and sludge was conducted in triplicates. The parameters such as pH, Ammonia-Nitrogen, Total organic carbon (TOC), soluble chemical oxygen demand (sCOD), and total volatile fatty acids (VFAs) of the mesophilic sludge were determined before starting the BMP test according to

American Public Health Association (APHA) standards to ensure quality (APHA, 2005). The proximate analyses of the samples (inoculum, PHB sample, cellulose) were also performed in accordance with the APHA standards. This protocol involves drying samples at 105 °C until they reach a constant weight to determine total solids (TS) content, followed by calcination at 550 °C for four hours to measure volatile solids (VS) content (Baird et al., 2017). The ultimate analysis was performed using a fully automated elemental analyzer (EURO Vector EA, Germany). The theoretical methane potential of all the samples used in this study was calculated from the stoichiometric equations (Eq. (2)) based on their elemental composition. The theoretical methane production (BMP_{thAtC}) was then calculated according to Eq. (3) (Nielfa et al., 2015).

$$C_nH_aO_bN_cS_d + \left(n - \frac{a}{4} - \frac{b}{2} + \frac{3c}{4} + \frac{d}{2}\right) H_2O \rightarrow \left(\frac{n}{2} + \frac{a}{8} - \frac{b}{4} + \frac{3c}{8} + \frac{d}{4}\right) CH_4 + \left(\frac{n}{2} - \frac{a}{8} + \frac{b}{4} + \frac{3c}{8} + \frac{d}{4}\right) CO_2 + c\ NH_3 + d\ H_2S \quad (2)$$

$$BMP_{thAtC} = \frac{22.4\left(\frac{n}{2} + \frac{a}{8} - \frac{b}{4} - \frac{3c}{8} - \frac{d}{4}\right)}{12n + a + 16b + 14c + 32d} \quad (3)$$

where n, a, b, c, and d represent the number of moles of C, H, O, N, and S, respectively.

2.4.3. Experimental setup

The BMP test is a preliminary study to investigate the methane production potential of organic materials. The tests were carried out in 500 mL reactors (serum bottles), which had 50 % of the working volume and left a headspace of 50 %. The inoculum-to-substrate ratio (ISR) for this study was set at 2.85 based on the optimal range for bioplastic (PHB) degradation suggested by (Cazaudehore et al., 2022). A total of 12 reactors were kept simultaneously for four groups (PHB filament, PHB sheet, Cellulose, and Blank in triplicates). All the serum bottles were filled with 250 mL inoculum, and the number of samples and cellulose required to be added in each reactor was calculated from the fixed ISR (Eq. (4)). The amounts of samples and cellulose needed to be added to 250 mL inoculum were obtained as 4 g for PHB filament. 3.87 g for PHB sheets and 3.79 g for cellulose.

$$ISR = \frac{\text{VS of inoculum (g)}}{\text{VS of substrate (g)}} \quad (4)$$

The sample reactors contained substrate and inoculum, whereas the blank bottle was filled with only inoculum without substrate, which was a reference to methane generation from the inoculum. Two silicone pipes with 5 mm internal diameter were fitted with the rubber cap of the reactors. One pipe was for attaching the Tedlar bags for gas collection, and the other was for nitrogen gas (N_2) purging, which was closely sealed during the test. N_2 was purged into each flask for approximately two minutes and closed air-tight with the caps and parafilm to provide

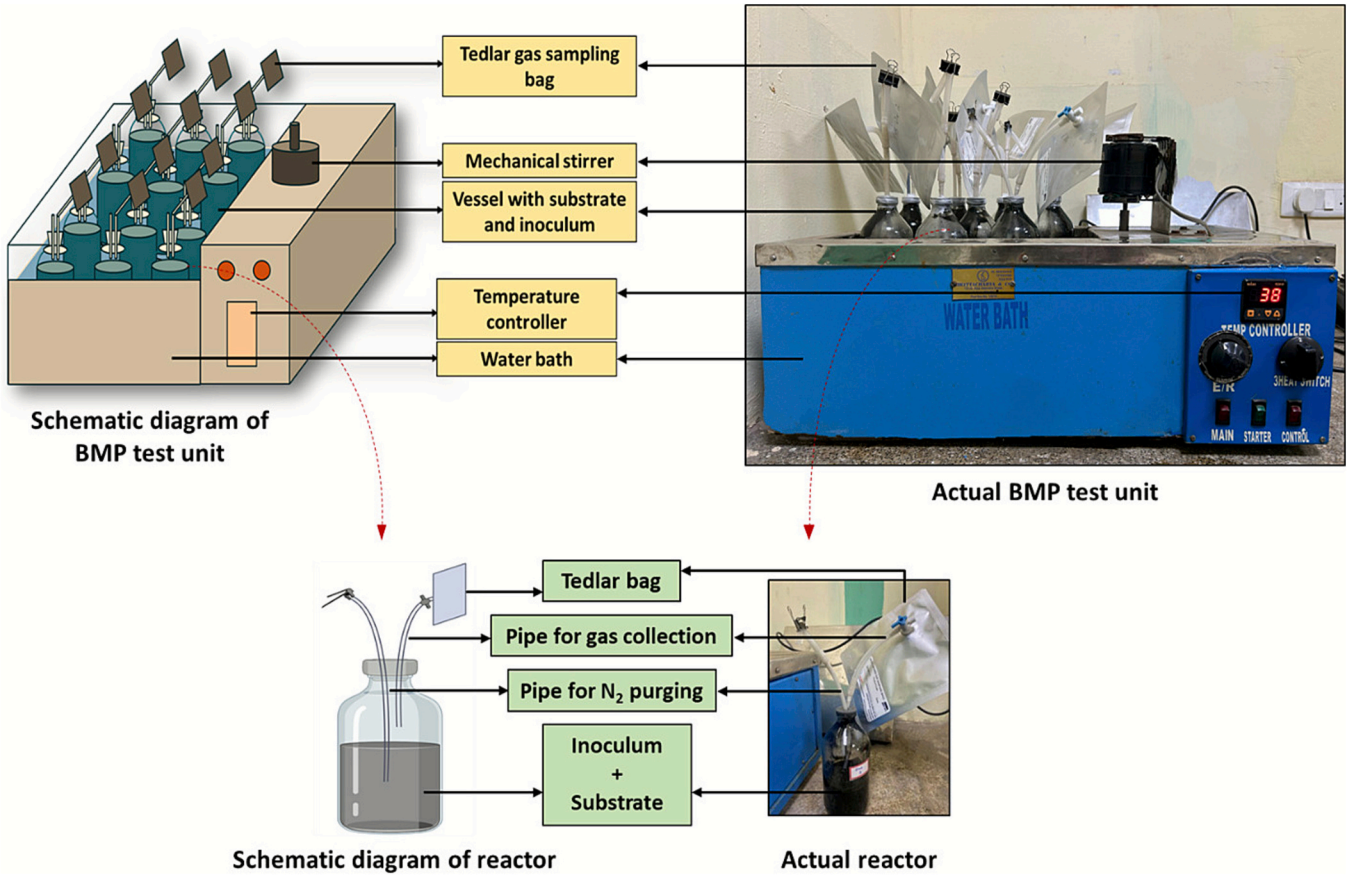


Fig. 2. Overall experimental setup and individual reactor utilized in this study.

an anaerobic environment. All bottles were placed in a water bath set at 38 °C (mesophilic condition). The reactors were manually lightly shaken daily for two minutes to ensure a homogeneous distribution of microorganisms, substrate, and temperature in serum bottles, as well as to help in the release of trapped gas bubbles in solution into the headspace of serum bottles. The incubation was planned to stop if the daily methane generation was less than 1 % over three consecutive days. The schematic diagram and the actual images of the BMP experimental setup, along with the individual rector used in this study, are shown in Fig. 2.

2.4.4. Gas volume measurement and composition analysis

The volume of gas collected in the Tedlar bags and the corresponding composition were analyzed daily. Gas volume measurements were conducted using the water displacement method. This involved an inverted graduated measuring cylinder clamped to a retort stand and immersed in a water bath, as illustrated in Fig. 3. The cylinder was initially filled with water. The Tedlar bag containing gas was securely connected to the inlet of the column using a silicone pipe with gas-tight connections to prevent leakage. The gas was then gently released into the column through a flow regulator, displacing water in the graduated cylinder. The volume of displaced water was recorded as the gas volume generated. Further, the composition was determined by injecting the gas with a 20 μL syringe into a Gas Chromatography-Mass Spectroscopy (GC–MS) instrument (Agilent 7890) Biogas production from the blank control (inoculum without test material) was subtracted from the biogas production of other samples to isolate test material contributions. All results were normalized to standard temperature and pressure conditions (P_{atm}, 0 °C).

Biodegradation of the samples was quantified by comparing the experimental methane production (BMP_{exp}) to the theoretical methane production (BMP_{thAtC}) calculated from elemental characterization (Eq. (5)).

$$\text{Biodegradation}\ (\%) = \frac{BMP_{exp}}{BMP_{thAtC}} \times 100 \quad (5)$$

Upon completing the experiment, the residual mixture of sludge and samples was characterized according to APHA standards to confirm the conversions during degradation (Baird et al., 2017).

3. Results and Discussion

3.1. Characterization of 3D printed PHB

3.1.1. FTIR

FTIR analysis was conducted to identify the functional groups in the filament and to verify its chemical composition by comparing it with previously documented spectra of PHB. The spectra, captured within the wavenumber range of 4000 to 600 cm^{-1} (as illustrated in Fig. 4), display

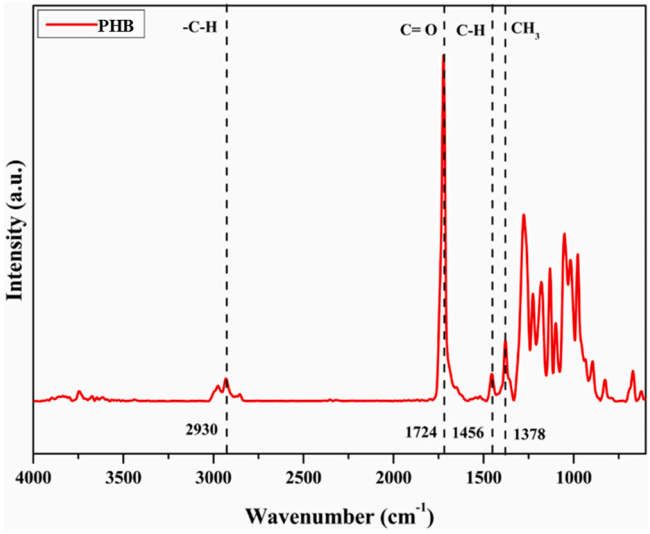


Fig. 4. FTIR spectroscopy peaks for PHB samples plotted in ATR mode.

prominent absorption peaks that correspond well with the distinctive vibrational modes of PHB. An acute and pronounced peak at 1724 cm^{-1} is attributed to the stretching vibrations of the carbonyl (C=O) functional group in ester bonds, a characteristic peak of PHB (Trakunjae et al., 2021). This peak aligns with previous research on PHB, validating the existence of ester linkages in its polymer backbone. A robust absorption band at 1277 cm^{-1} is ascribed to -CH (alkyl) bending vibrations, further corroborating the polymer's structural integrity (Randriamanhefa et al., 2003). A distinct peak at 2930 cm^{-1} is attributed to C—H stretching of alkanes, often present in PHB and its derivatives (Hamdy et al., 2022). Additionally, the absorption band ranging from 1010 cm^{-1} to 1300 cm^{-1} is attributed to C—O stretching vibrations, a well-established characteristic of PHB and other polyesters (Chan et al., 2011). The detected FTIR peaks show a significant agreement with previously documented spectra of PHB, indicating peak consistency and structural resemblance (Wu et al., 2001). Based on reported detection limits for ATR-FTIR in polymer matrices, additives with strong, distinct functional groups are generally detectable at concentrations of approximately ≥1 wt% under standard conditions, provided their absorption bands do not overlap with those of the base polymer (Cuthbertson et al., 2024). While trace components below this threshold may not be identified by FTIR alone, none were intentionally added, and the supplier confirmed the filament as pure PHB. Moreover, as we are comparing the spectra between two processing forms, i.e., the filament and the 3D-printed sheet, even low-level traces of additives below 1 wt% would not affect the comparative biodegradation results.

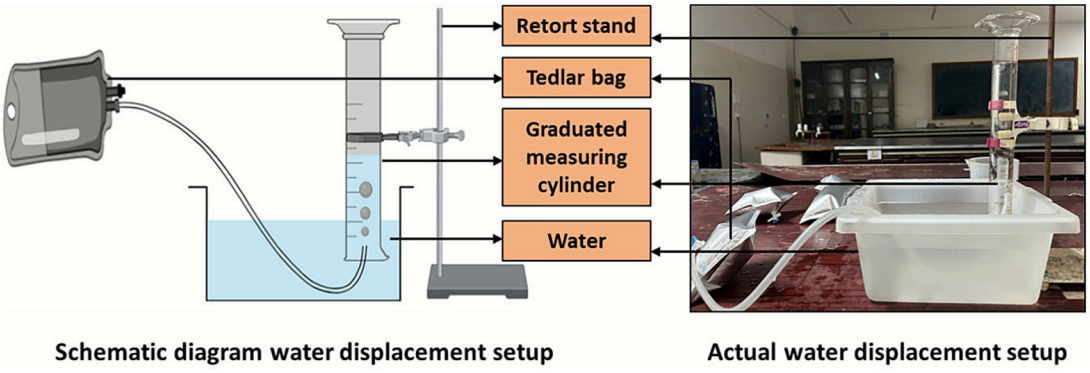


Fig. 3. Water displacement setup used for gas volume measurement

3.1.2. XRD

PHB, a well-known member of the polyhydroxyalkanoate (PHA) family, exhibits a semi-crystalline nature with characteristic diffraction peaks in XRD analysis. The primary crystalline reflections of PHB are observed at 2θ values of approximately 13.6°, 16.9°, and 22.2°, corresponding to the (020), (110), and (021) crystallographic planes, respectively, which confirm its orthorhombic crystal structure (Pradhan et al., 2018). These diffraction patterns are well-documented for PHB and its copolymers, indicating a highly ordered molecular arrangement that influences its mechanical and thermal properties (Barham et al., 1984). Additional diffraction peaks observed at 26.9°, 28.8°, 37.8°, 44.14°, 48.9°, 64.20°, 70.35°, and 77.27° arise from a combination of crystallographic, processing-related, and extrinsic factors. The peaks correspond to higher-order Miller indices (e.g., (040), (220)) of PHB's orthorhombic lattice, identifiable due to increased crystallite coherence, along with processing-induced effects like annealing, which modifies crystallization kinetics by facilitating chain reorganization into metastable phases or altering lamellar thickness. Thermal treatments above the T_g done during the filament production as well as 3D printing, enhance secondary crystallization, but contaminants (such as residual catalysts and nucleating agents) or secondary phases can also add foreign crystalline signals. Mechanical processing can create residual stress, which usually presents as peak broadening instead of distinct reflections. The identified higher-angle characteristics correspond with research associating their formation with structural rearrangement during thermal treatment, since annealing improves long-range order and intensifies higher-order reflections (Barham et al., 1984; Zhang et al., 2024b) (Fig. 5).

The calculated crystallinity percentage is approximately 40.9 % for the filament, increasing to around 61.8 % after FDM 3D printing. The enhancement in crystallinity of the FDM-printed component, as inferred from XRD analysis, is attributable to the thermal and mechanical dynamics of the FDM process. During the production of the PHB filament through pultrusion/extrusion, it experiences fast cooling, which kinetically entraps polymer chains in disordered amorphous areas, hence constraining crystallinity. In FDM printing, the PHB filament is heated to a semi-molten state (about 190–210 °C), facilitating chain mobility and rearrangement. Shear forces are exerted in the nozzle-oriented polymer chains along the deposition axis, facilitating crystalline domain nucleation. After the deposition of layers in FDM, reduced cooling rates relative to filament production provide prolonged periods for lamellar stacking into organized planes, enhancing long-range crystallographic order. This corresponds with the obtained XRD results, which exhibit stronger peaks after printing, signifying improved crystallite coherence. Moreover, the partial melting of the filament's metastable crystalline areas during FDM facilitates recrystallization into thermodynamically advantageous configurations, hence enhancing crystallinity. This behaviour aligns with PHB's tendency for secondary crystallization after thermal annealing, as shown in previous research (De Lima Souza et al., 2009; Di Lorenzo and Androsch, 2019).

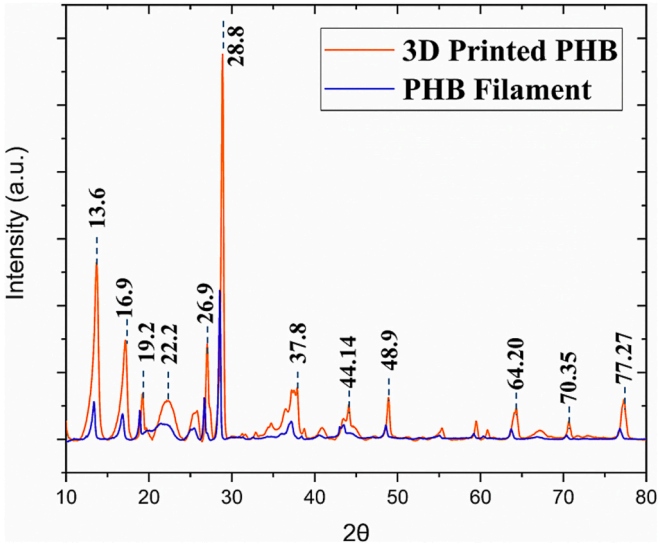


Fig. 5. X-ray diffraction peaks for PHB filament and PHB 3D printed samples.

Table 3
Properties of the inoculum, cellulose, PHB filament, and PHB sheets (3DP).

Parameters	Unit	Mesophilic inoculum	PHB sheet (3DP)	PHB Filament	Cellulose
pH		7.22 ± 0.05			
Ammonia-Nitrogen	gN-NH_3.L^{-1}	0.63 ± 0.07			
TOC	g.L^{-1}	0.45 ± 0.01			
sCOD	g.L^{-1}	426 ± 184			
Total VFAs	g.L^{-1}	0.34 ± 0.01			
Total Solids	% raw mass	8.02 ± 0.06	97.4 ± 0.37	98.13 ± 0.13	95.21 ± 0.01
Volatile Solids	% raw mass	4.04 ± 0.1	91.72 ± 0.23	88.72 ± 0.09	94.81 ± 0.18
Carbon	% TS	28.25 ± 1.73	50.37 ± 0.04	50.43 ± 0.09	43.35 ± 1.41
Hydrogen	% TS	3.51 ± 0.36	6.88 ± 0.07	6.53 ± 0.09	6.49 ± 0.27
Nitrogen	% TS	4.13 ± 0.18	1.14 ± 0.01	1.15 ± 0.01	0 ± 0.00
Sulfur	% TS	2.95 ± 0.09	0 ± 0.00	0 ± 0.00	0 ± 0.00
Oxygen	% TS	61.17	37.85	38.13	44.97
Theoretical methane potential	mL CH_4/g VS	94.32	435.12	426.57	361.92

Data presented as mean values ± standard deviation ($n = 3$).

3.2. Biodegradation of PHB filament and its 3D printed sheet under mesophilic conditions

Table 3 details the physicochemical properties of both the inoculum and the substrates. The inoculum characteristics complied with the quality standards defined by Holliger et al. (2016) (Holliger et al., 2021).

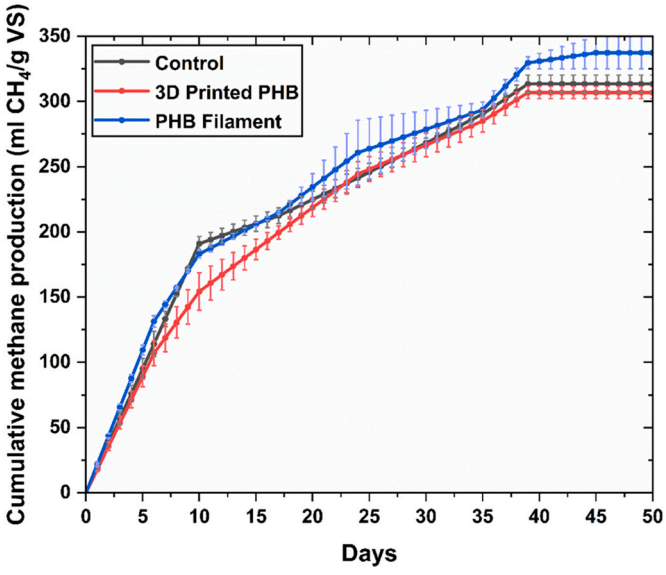


Fig. 6. The mean cumulative methane production (mL CH_4/g VS) from the PHB filament and PHB sheet under mesophilic conditions.

The cumulative methane production obtained from the anaerobic digestion of PHB filament and 3D printed PHB sheet under mesophilic conditions is demonstrated in Fig. 6. The mesophilic BMP assay (37 °C, ISR = 2.85) resulted in 86.3 % biodegradation of the microcrystalline cellulose powder (positive control) after 42 days, thereby validating inoculum activity and test integrity (Yagi et al., 2014). Under identical conditions, PHB filaments showed rapid methane evolution, reaching a cumulative biodegradation of 78.24 % by day 49. In contrast, the 3D-printed sheet achieved 70.35 % biodegradation within 42 days, demonstrating that both samples possess high biodegradability. The biodegradation plateau for the filament coincides with mesophilic anaerobic studies, indicating that PHB films degrade to roughly 80 % within 45 to 60 days (Kim et al., 2023). The 8 % delay observed in the 3D printed sample is directly related to the variation in crystallinity (40.9 % for the filament and 61.8 % for the sheet) discussed in the preceding section. During anaerobic digestion, PHB undergoes ester bond cleavage, releasing soluble intermediates such as 3-hydroxybutyric acid, dimers, and short oligomers that are subsequently metabolized to methane and CO_2. These hydrolysis products are identical for both the filament and printed forms; however, the increased crystallinity of the printed PHB reduces chain mobility and accessibility of ester bonds, making the material more resistant to cleavage rather than altering the degradation pathway itself. Filament extrusion and layer-by-layer deposition in fused filament fabrication provide enhanced chain alignment and recrystallization, hence reducing the amorphous regions vulnerable to hydrolytic cleavage (Kanabenja et al., 2024). Higher crystalline fractions restrict water infiltration and enzyme accessibility, hence retarding the hydrolysis phase that precedes microbial assimilation (Paloyan et al., 2025). In mesophilic conditions, amorphous domains are recognized to deteriorate first, resulting in a crystalline framework that dissolves progressively (Bher et al., 2022). Furthermore, the dense layer interfaces and lower porosity of the 3D-printed sheet may restrict microbial colonization sites and hinder the mass transfer of breakdown by-products, hence obstructing overall biodegradation (Herrera et al., 2024). Conversely, melt-cast or injection-molded PHB typically exhibits reduced crystallinity and increased surface roughness, promoting accelerated microbial adhesion and enzymatic activity (Kim et al., 2023). FDM 3D printing induces shear alignment and slower cooling, promoting lamellar stacking and elevating crystallinity to 61.8 % in the printed part. The dense orthorhombic lattice of crystalline regions sterically hinders enzyme penetration and reduces chain mobility, delaying hydrolysis and methane production in biomethane potential (BMP) tests (Thapliyal and Chandra, 1990). This aligns with studies demonstrating that higher crystallinity in PHB correlates with reduced microbial degradation rates in anaerobic environments, as methanogens depend on hydrolyzed monomers from amorphous zones (Tripathi et al., 2013). Thus, the filament's lower crystallinity facilitates faster biodegradation, while 3D-printed PHB's enhanced crystallinity imposes kinetic barriers to enzymatic depolymerization. While the observed 8 % reduction in biodegradability and 10 % lower methane yield for the 3D-printed sheet may not significantly hinder industrial anaerobic digestion performance, our BMP study was designed to capture even subtle variations that reflect processing-induced changes in polymer structure. These insights are valuable for optimizing 3D printing parameters to ensure consistent biomethane recovery and material breakdown in circular waste-to-energy systems. By linking crystallinity changes to BMP outcomes, we provide a scientific basis for improving both product performance and end-of-life energy recovery efficiency. Both the filament and the 3D-printed sheet exceeded the 60 % biodegradation threshold described by relevant ASTM/ISO criteria within the standard test duration, thereby confirming that both materials are biodegradable under mesophilic anaerobic conditions (Meereboer et al., 2020). Research on PHBV films demonstrated that semi-crystalline materials degrade up to 40 % more slowly than fully amorphous counterparts under analogous anaerobic-sludge conditions (Steiner et al., 2022). Similarly, the hydrolysis rates mediated by depolymerase in PHB significantly decline as the average crystal size and overall crystallinity increase, with rate constants decreasing by 30–50 % when crystallinity rises from approximately 35 % to 65 % (Sarobe et al., 1998).

Table 4
Final characterization of the degraded samples.

Parameters	Unit	Mesophilic sludge	Blank	Control	Sheet	Wire
pH		7.22 ± 0.05	8.45 ± 0.05	8.44 ± 0.04	8.39 ± 0.03	8.45 ± 0.04
TOC	$g.L^{-1}$	0.45 ± 0.01	0.61 ± 0.04	0.82 ± 0.03	0.72 ± 0.06	0.75 ± 0.04
sCOD	$mg.L^{-1}$	426 ± 184	1624 ± 143	1268 ± 157	2316 ± 15	2185 ± 46
Total VFAs	$g.L^{-1}$	0.34 ± 0.01	0.19 ± 0.02	0.21 ± 0.02	0.24 ± 0.01	0.26 ± 0.01
Volatile Solids	% raw mass	4.04 ± 0.1	4.34 ± 0.11	2.97 ± 0.08	3.16 ± 0.17	3 ± 0.16
Carbon	% TS	28.25 ± 1.73	27.58 ± 0.45	27.04 ± 0.76	28.02 ± 1.23	29.21 ± 1.24
Hydrogen	% TS	3.51 ± 0.36	3.44 ± 0.14	3.59 ± 0.05	3.7 ± 0.07	3.71 ± 0.11
Nitrogen	% TS	4.13 ± 0.18	3.82 ± 0.49	3.57 ± 0.07	3.56 ± 0.46	3.95 ± 0.17
Sulfur	% TS	2.95 ± 0.09	2.18 ± 0.55	1.95 ± 0.50	2.13 ± 1.14	2.57 ± 0.16
Oxygen	% TS	61.17	62.98	63.85	62.59	60.56

Data presented as mean values ± standard deviation (n = 3).

The physicochemical characteristics of the digestate from all the reactors were analyzed after completion of the test and the results are presented in Table 4. The initial inoculum pH of 7.22 increased to 8.39 in the sheet digester and 8.45 in the filament reactor towards the completion of the test. The ammonification of proteinaceous material releases NH_4^+, which causes alkaline shifts that are typical transformations of mesophilic AD. Simultaneous bicarbonate buffering also raises pH towards the mild alkaline range (pH 8.0–8.5) without preventing methanogenesis (Chen et al., 2024). In addition to protecting against acid spikes during hydrolysis and acidogenesis, this buffering ability keeps methanogen conditions stable. The TOC of the inoculum (0.45 g/L) increased to 0.72 g/L with PHB sheets and 0.75 g/L with PHB filaments, signifying a 60 % and 67 % increase in TOC, respectively. TOC values were interpreted as indicators of soluble organic carbon released during hydrolysis rather than as a direct measure of microbial biomass. The increase in TOC thus reflects the conversion of PHB into soluble degradation intermediates, supporting the progression of the anaerobic biodegradation process. In similar mesophilic digesters, the amount of TOC released from complex substrates (such as lignocellulosic or food-waste slurries) usually ranged from 44 to 76 % over 20 to 40 days (Yan et al., 2020). The slight lower TOC accumulation in the sheet digesters reflects their slower hydrolysis kinetics due to higher crystallinity. This comparison is intended solely as an order-of-magnitude benchmark rather than a claim that PHB behaves identically to complex wastes. In this study, TOC increased by 60–67 % during mesophilic BMP testing, while literature reports (Yan et al., 2020) TOC increases of 44–76 % over 20–40 days in digesters fed with lignocellulosic or food-waste substrates. Although PHB is a chemically uniform polyester rather than a heterogeneous waste, the hydrolysis–acidogenesis–methanogenesis pathway is common to both systems. The magnitude of TOC release therefore provides a valid indicator of conversion efficiency, demonstrating that PHB under mesophilic conditions undergoes substantial solubilization comparable to other biodegradable organics. This comparison is presented only to show that the extent of carbon release from PHB lies within the same range as other substrates under similar BMP conditions, while acknowledging differences in substrate composition, inoculum activity, and operating parameters.

The sCOD increased dramatically from an initial 426 mg/L to 2316

mg/L (sheet) and 2185 mg/L (filament). This increase underscores extensive solubilization of PHB macromolecules into low-molecular-weight organics. During acidogenic conditions, sCOD generation is strongly correlated with VFA accumulation, with studies showing linear sCOD–VFA relationships and optimal hydrolysis at pH 4.5–5.0 (Gomes et al., 2016). Although the AD system in this study remained above pH 7.8, the magnitude of sCOD elevation reflects robust hydrolytic activity. This aligns with previous reports of hydrolytic bacteria converting more than 50 % of polymeric VS to sCOD in two weeks under mesophilic BMP conditions. VFAs declined from an initial 0.34 g/L to 0.24 g/L (sheet) and 0.26 g/L (filament). VFAs declined from an initial 0.34 g/L to 0.24 g/L (sheet) and 0.26 g/L (filament). This reduction indicates efficient acidogenesis and subsequent acetogenesis, wherein acidogenic bacteria ferment hydrolysis products into VFAs (acetate, propionate, butyrate) and methanogens then consume these intermediates (Laiq Ur Rehman et al., 2019). The slightly higher residual VFA in the filament digesters aligns with their faster hydrolysis and thus greater VFA generation, which was then gradually consumed during methanogenesis. Summarizing, the observed changes in pH, TOC, sCOD, and VFA profiles together reflect the behaviour of conventional mesophilic AD. The delayed biodegradation and associated physicochemical changes for the 3D-printed sheet compared to the filament are associated with the elevated crystallinity, which hinders water uptake, enzyme penetration, and microbial colonization, as reported by prior studies on anaerobic digestion of crystalline PHB and its copolymers.

4. Conclusion

This study evaluated the anaerobic biodegradability of PHB in filament form and after FDM 3D printing into sheets using the standardised BMP test (ASTM D5511–11). FTIR analysis confirmed that the chemical composition of both forms was PHB, while XRD revealed a substantial increase in crystallinity in the 3D-printed sheet compared to the filament. Higher crystallinity reduces chain segment mobility, decreases free volume, and limits the diffusion of water and enzymes into the polymer matrix, which in turn slows hydrolytic chain scission and subsequent microbial assimilation. Under mesophilic BMP conditions, this structural difference resulted in a measurable reduction in biodegradation extent and methane yield for the printed sheet relative to the filament.

The observed trends in pH, TOC, sCOD, and VFAs further support this interpretation, indicating more rapid hydrolysis and substrate conversion for the less crystalline filament. These results highlight that processing-induced crystallinity changes must be accounted for when evaluating the environmental fate of biodegradable polymers. For additive manufacturing applications, understanding and controlling the relation between processing parameters, resulting microstructure, and biodegradation kinetics is essential for optimizing both material performance and sustainability outcomes.

CRediT authorship contribution statement

Anita Jena: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Anjaly P. Thomas:** Writing – review & editing, Visualization, Validation, Methodology. **Bijaya Bikram Samal:** Writing – review & editing, Visualization, Supervision, Software, Methodology, Formal analysis, Conceptualization. **Brajesh Kumar Dubey:** Writing – review & editing, Supervision, Resources, Project administration. **Cheruvu Siva Kumar:** Writing – review & editing, Supervision, Resources, Project administration. **Shailendra Kumar Varshney:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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